



UNIVERSITY OF KELANIYA
SOFTWARE ENGINEERING TEACHING UNIT
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SENG 31333 – Business Intelligence with Power BI
Lab – In-Class Activity 02

This session will guide you through the development of a real-world Power BI analytics project by designing and building a **Food Waste Monitoring Dashboard for Restaurants**. Students will work through the complete business intelligence lifecycle, including data preparation, cleaning messy operational data, creating relationships, building DAX measures, designing interactive dashboards, and generating business insights.

Business Scenario

A restaurant chain with multiple branches is experiencing rising food waste, resulting in unnecessary costs and reduced profits.

Currently, waste data is manually tracked in Excel, and management cannot clearly identify:

- Which food items are wasted the most
- Which branches are inefficient
- What are the reasons that cause waste
- Which time periods create the highest waste
- How future waste can be reduced

You are hired as a Data Analyst to build a Power BI solution.

Required Power BI Dashboard Pages

Page 1: Executive Overview

Cards

1. Total Waste Quantity
2. Total Waste Cost
3. Waste % vs Sales
4. Worst Performing Branch
5. Most Wasted Item

Charts

1. Waste Cost Trend by Month
2. Waste by Branch
3. Waste by Category

Page 2: Root Cause Analysis

Visuals

1. Waste by Reason (Expired / Overcooked / Customer Return / Spillage)
2. Waste by Shift
3. Waste by Day of Week
4. Top 5 Most Wasted Items

Page 3: Profit Impact Dashboard

Measures

1. Potential Revenue Lost
2. Monthly Savings Opportunity

Page 4: Forecasting & Planning

Visual: (Use Analytics Pane Forecast)

1. Next 30 days waste prediction.

Important DAX Measures

Total Waste Quantity

Total Waste Qty = SUM(WasteData[Qty Wasted])

Total Waste Cost

Waste Cost = SUMX(WasteData, WasteData[Qty Wasted] * WasteData[Unit Cost])

Total Revenue

Total Revenue = SUM(SalesData[Revenue])

Waste %

Waste % = DIVIDE([Total Waste Cost], [Total Revenue], 0)

Worst Performing Branch

Worst Branch = TOPN(
1,
VALUES(BranchData[Branch]),
[Total Waste Cost],
DESC)

Most Wasted Item

Most Wasted Item =

TOPN(

1,

VALUES(ItemMaster[Item Name]),

[Total Waste Qty],

DESC

)

Monthly Savings opportunity

Savings Opportunity = [Total Waste Cost] * 0.20

Potential Revenue Lost

Potential Revenue Lost = [Total Waste Cost]

Step-by-Step Guidelines

1. Part A – Data Loading

Import all data (Excel files and CSV files, load one by one) into Power BI by identifying the data structure.

- Check row counts
- Identify bad columns
- Detect blanks

2. Part B – Power Query Cleaning

Note: Why Clean Data?

If you don't clean in Power Query, you're not doing "data analysis", you're just decorating messy data with charts.

Checklist:

- ☒ Correct data types
- ☒ Text trimmed & cleaned
- ☒ Missing values handled
- ☒ Remove / Replace null rows
- ☒ Remove duplicates
- ☒ Rename columns properly
- ☒ Categories standardized (Branch names)

Transform the data in the Power Query Editor and review each table systematically using a defined checklist to clean the dataset. Start with the 'WasteData' table, then apply the same process consistently to the remaining four tables by validating and refining each column step by step.

Date Column — Use the column filter option to explore the data and understand its structure.

1. Change the data type to Date.
2. Use the column filter option to explore the data and understand its structure. What are the observations?

	Date	Branch	ABC 123 Item Name	ABC 123 Category
1	1/15/2025	A	Rice Bowl	Main Meal
2	5/15/2025	A		
3	6/10/2025	A		
4	5/29/2025	A		
5	1/9/2025	A		
6	3/7/2025	A		
7	1/4/2025	A		
8	4/20/2025	A		
9	6/10/2025	A		
10	3/15/2025	A		
11	3/9/2025	A		
12	4/16/2025	A		
13	5/21/2025	A	Coffee	Beverage
14	5/25/2025	A	Fries	Snacks
15	5/28/2025	A	Coffee	Beverage
16	1/20/2025	Colombo	Rice Bowl	Main Meal

Remove Top Rows

Specify how many rows to remove from the top.

Number of rows

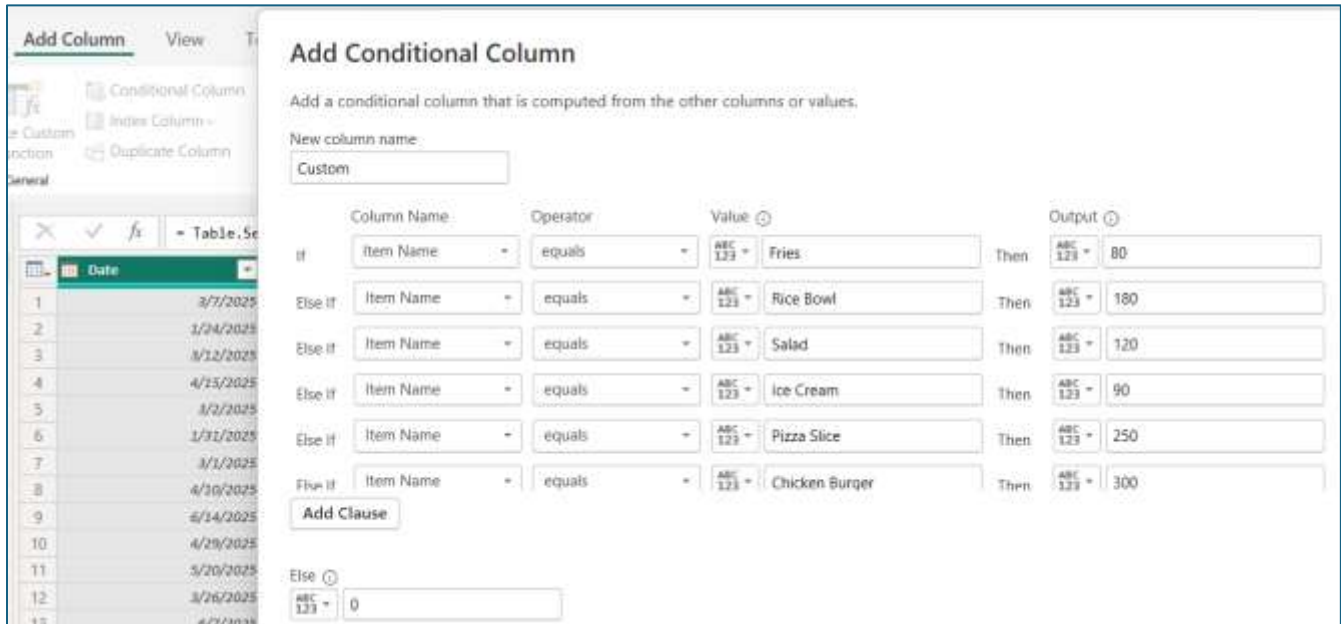
The screenshot shows the Power BI Desktop interface with a table named 'Branch' containing 15 rows of 'Colombo'. The 'Transform' menu is open, and the 'Transform' option is selected, which has opened a sub-menu. The sub-menu contains the following options: 'lowercase', 'UPPERCASE', 'Capitalize Each Word', 'Trim', 'Clean', 'Length', 'JSON', and 'XML'. The 'Capitalize Each Word' option is highlighted in the sub-menu.

Item Name, Category, Waste Reason, Shift and Quantity Wasted Columns

1. Change the data type to Text. / Quantity wasted to Whole Number
2. Use the column filter option to explore the data and understand its structure. What are the observations?
 - Nothing to change

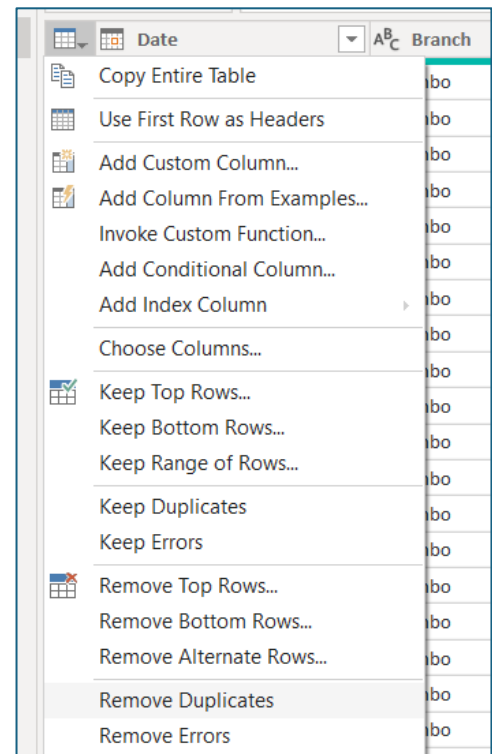
Unit Cost Column

1. Change the data type to Fixed Decimal Number (also known as **Currency**).
2. Use the column filter option to explore the data and understand its structure. What are the observations?



After cleaning all columns, remove duplicate records from the entire table by identifying and eliminating identical values across all relevant columns.

1. Open **Transform Data**.
2. Click the **table icon** (top-left corner of the data grid).
3. Select **Remove Duplicates**.



Now the WasteData table cleaning is completed. Apply the same cleaning process consistently to the remaining four tables by validating and refining each column step by step.

SalesData Table

1. Change data types (e.g., set Revenue to Decimal Number).
2. Split and reformat the Date column in the same way as the WasteData table.
3. Revenue = money earned from sales. So logically Revenue should NEVER be negative. Check it and validate.

The screenshot shows a data tool interface. On the left, a filter menu is open with options: Sort Ascending, Sort Descending, Clear Sort, Clear Filter, Remove Empty, and Number Filters. Below these is a search bar and a list of values: (Select All), 3300, 3900, 4050, 4140, 4800, and 5400. A dropdown menu is open over the list, showing comparison operators: Equals..., Does Not Equal..., Greater Than..., Greater Than Or Equal To..., Less Than..., Less Than Or Equal To..., and Between... On the right, the 'Filter Rows' panel is visible. It has a title 'Filter Rows' and a subtitle 'Apply one or more filter conditions to the rows in this table.' Below this are two radio buttons: 'Basic' (selected) and 'Advanced'. The main section is titled 'Keep rows where 'Revenue'' and contains two filter conditions. The first condition is 'is less than' followed by a dropdown menu showing '0'. The second condition is 'And' followed by a dropdown menu showing 'Enter or select a valu...'. There is also an 'Or' radio button.

Calender Table

1. Split and reformat the Date column in the same way as the WasteData table.

ItemMaster Table

1. Change data types (costs into Fixed Decimal Number).

3. Part C – Data Modeling

Note: Why Data Modeling?

Raw tables are not designed for analysis. Modeling structures raw data into a consistent and efficient format so that relationships, calculations, and analysis produce accurate, reliable, and scalable results.

See the current model in Model view. What are the observations?

Use Star Schema

- Clean structure
- Easy filtering
- Accurate measures
- Fast queries

Fact Tables: WasteData, SalesData

Dimension Tables: BranchData, Calendar, ItemMaster

Manage relationships ✕				
+ New relationship		⚡ Autodetect	Edit Delete Filter	
☐ From: table (column) ↑	Relationship	To: table (column)	Status	
☐ SalesData (Branch)	* — ◀ — 1	BranchData (Branch)	Active	...
☐ SalesData (Date)	* — ◀ — 1	Calendar (Date)	Active	...
☐ SalesData (Item Name)	* — ◀ — 1	ItemMaster (Item Name)	Active	...
☐ WasteData (Branch)	* — ◀ — 1	BranchData (Branch)	Active	...
☐ WasteData (Date)	* — ◀ — 1	Calendar (Date)	Active	...
☐ WasteData (Item Name)	* — ◀ — 1	ItemMaster (Item Name)	Active	...

4. Part D – DAX Measures

Note: Why DAX Measures?

DAX measures exist because Power BI is not supposed to store calculations inside tables, it is supposed to compute answers dynamically based on context.

Create a new table as 'MeasureTable' to include all the Measures there.

5. Part E – Dashboard Visualization

Create four pages as mentioned to visualize insights.

Page 1: Executive Overview

Cards

- Total Waste Quantity
- Total Waste Cost
- Waste % vs Sales
- Worst Performing Branch
- Most Wasted Item

Charts

- Waste Cost Trend by Month
- Waste by Branch
- Waste by Category

Food Waste Monitoring Dashboard - 2025

Executive Overview

3K

Total Waste Qty

540K

Total Waste Cost

\$0.0482

Waste %

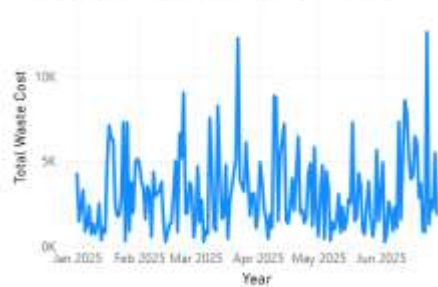
Colombo

Worst Branch

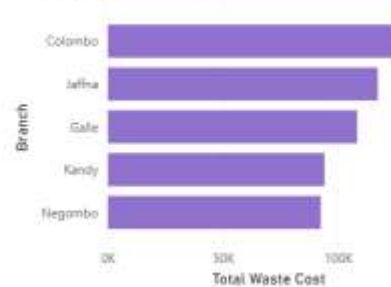
Fries

Most Wasted Item

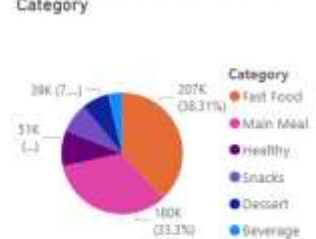
Total Waste Cost by Year, Quarter, Month and Day



Total Waste Cost by Branch



Total Waste Cost by Category and Category

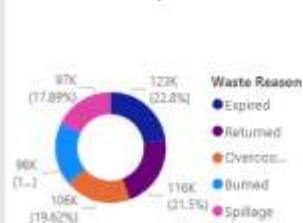


Page 2: Root Cause Analysis

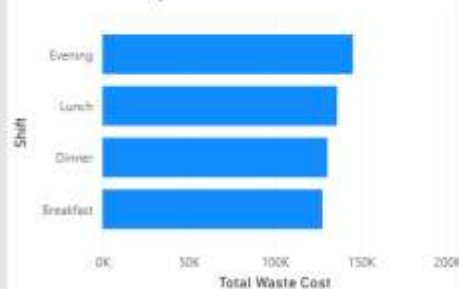
- Waste by Reason (Expired / Overcooked / Customer Return / Spillage)
- Waste by Shift
- Waste by Day of Week (New column in WasteData; Day Name = FORMAT(WasteData[Date], "dddd")
- Top 5 Most Wasted Items

Root Cause Analysis

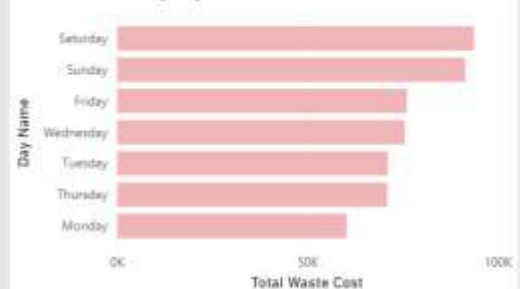
Total Waste Cost by Waste Reason



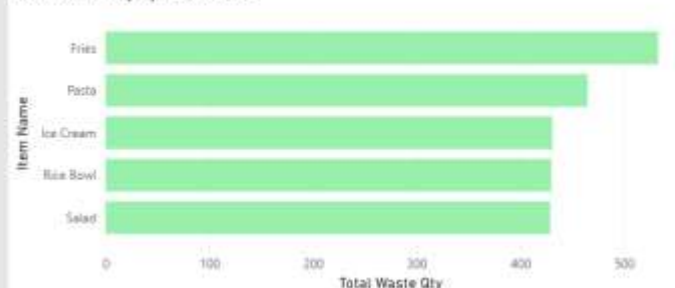
Total Waste Cost by Shift



Total Waste Cost by Day Name

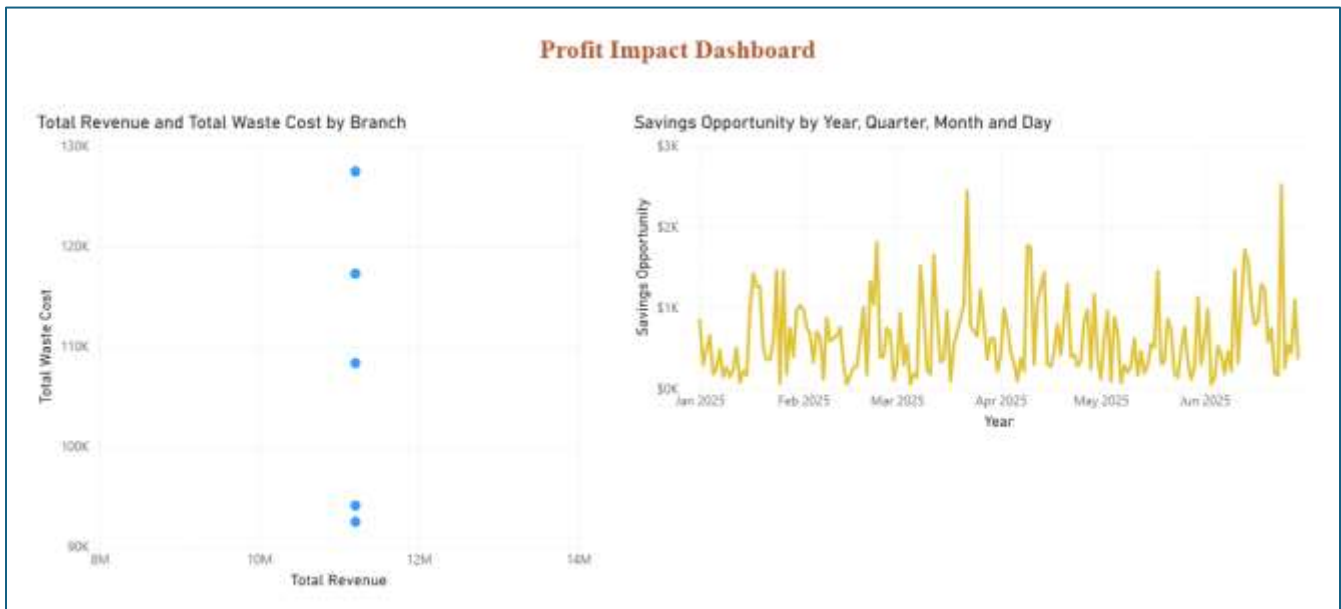


Total Waste Qty by Item Name



Page 3: Profit Impact Dashboard

- i. Potential Revenue Lost
- ii. Monthly Savings Opportunity



Page 4: Forecasting & Planning

Visual: (Use Analytics Pane Forecast)

- i. Next 30 days waste prediction.

